Scarcity Is Not Enough: Structural Limits of Linear Sybil Cost Under Parallelizable Resources

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Permissionless systems resist Sybil attacks by binding influence to scarce resources. Yet influence concentration persists across systems built on computation, capital, and other reusable resources despite substantial differences in protocol design. This raises a fundamental question: is influence concentration primarily a consequence of protocol rules, or of the structural properties of the underlying resource itself?

In this paper, we address this question from a resource-centric perspective through the adversarial cost function $C(s,T)$, defined as the minimum expenditure required to sustain influence equivalent to controlling s independent participants over a time horizon of length T. Using this framework, we develop an axiomatic resource taxonomy that connects resource structure to adversarial cost scaling.

Our analysis shows that resource-mechanism pairs satisfying divisibility, additivity of influence, temporal reusability, and identity transferability admit influence amortization, yielding $C(s,T) = o(sT)$. Conversely, throughput-bounded, non-transferable, window-local resources enforce $C(s,T) = \Omega(sT)$, with marginal cost $\Delta(s,T) = \Omega(T)$ increasing with the time horizon. Together, these results reveal a fundamental asymptotic separation between resource classes that admit influence amortization and those that enforce linear cost.

These results shift the focus of decentralization from protocol design to resource design. Within mechanisms preserving the relevant parallelizability properties, decentralization limits arise from resource structure rather than consensus rules alone. The same properties also enable concentrated control to be projected across many nominally distinct participants through delegation, pooling, or identity replication, obscuring the relationship between visible identities and underlying control. The search for stronger decentralization must therefore begin with the choice of the underlying resource.

CCS Concepts: • **Security and privacy** → *Pseudonymity, anonymity and untraceability*; *Digital rights management*; • **Networks** → Peer-to-peer protocols.

Additional Key Words and Phrases: Sybil resistance, influence concentration, parallelizable resources, throughput-bounded resources, adversarial cost scaling, structural limits, resource taxonomy, permissionless systems

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1 INTRODUCTION

Permissionless systems rest on a radical premise: that open participation, without gatekeepers or trusted authorities, can produce secure and fair collective outcomes. The mechanism that makes this possible—binding influence to the expenditure of a scarce resource—has been the cornerstone of distributed consensus since Nakamoto [32]. Yet deployment has revealed a persistent gap between this promise and reality. Bitcoin's hash power is concentrated in a handful of mining pools [17]. Ethereum's stake is dominated by a small number of liquid staking providers [38].

This paper identifies a structural explanation for this phenomenon.

1.1 Resource-Based Defenses and Their Shared Blind Spot

The foundational obstacle in permissionless systems is the Sybil attack [12]: a single adversary generates many identities to amplify influence beyond any honest participant. The standard response is to make identity costly by tying participation to a scarce resource. Proof-of-Work binds influence to computational effort [16, 32] (specifically, to mining hardware capacity as a reusable stock, and energy as a per-window flow). Proof-of-Stake binds it to financial capital [24]. Graph-based defenses [46, 47] constrain identity placement via social structure. Identity-binding mechanisms [3, 15] restrict credential issuance to verified participants.

Despite their diversity, these approaches share a common blind spot. The original insight behind resource-weighted participation—introduced by Proof-of-Work—was precise: binding influence to a scarce resource prevents cheap identity creation from amplifying adversarial power. Subsequent work has studied this principle extensively, asking whether an adversary's *aggregate* resource share crosses a safety threshold.

What has received far less attention is a different question: does the *structure* of that resource permit influence to be concentrated at sublinear cost? A resource that is divisible, transferable, and reusable allows a stock of size $O(s)$ to sustain s-unit influence over an arbitrarily long horizon without any T-proportional renewal cost. The critical issue is therefore not merely scarcity, but whether influence can be amortized over time.

More importantly, the same resource properties allow concentrated control to be projected across many nominally distinct participants. A mining pool, for example, may aggregate a large concentration of hash power while distributing block-production opportunities across thousands of visible miners. Similar effects arise through delegation and other coordination mechanisms, where a single underlying concentration of resources can appear as a diverse population of independent actors. The challenge is therefore not only influence concentration, but the growing disconnect between visible participation and effective control.

This paper characterizes exactly which resource properties permit this temporal amortization, and which ones prevent it.

1.2 Why Concentration Emerges as a Structural Consequence

The concentration patterns observed across deployed systems are often discussed as protocol-level phenomena. Our framework suggests that they may instead arise from the structural properties of the resources on which those protocols are built.

In Bitcoin, mining pools aggregate hash power from thousands of participants into a small number of coordinated entities. Gencer et al [17] document that the top-four pools routinely control more than half of the network's total hash power. The reason may be structural rather than purely procedural. Mining hardware capacity—the reusable stock component of Proof-of-Work—can be accumulated, redistributed, and reused over time. A sufficiently large hardware base can therefore sustain influence indefinitely, while being allocated across many miners, pools, or operational identities. The per-block energy cost is a separate window-local flow component and is analyzed independently.

Proof-of-Stake replaced computational effort with financial capital, eliminating the window-local energy flow of PoW while retaining a similar stock structure. Financial capital can likewise be accumulated, redistributed, and reused over time. Delegation mechanisms allow a concentrated capital base to support influence across many validator identities without requiring proportional renewal costs in each participation window [24, 43]. The outcome is familiar: as of 2023, Lido Finance alone controlled approximately thirty percent of all staked ETH [38].

The same structural argument extends beyond PoW and PoS. Whenever a resource can be accumulated, redistributed, and reused, a concentrated resource base can sustain influence over an arbitrarily long horizon at cost that grows far more slowly than the duration of participation. Moreover, the same concentration of resources can often be projected through many nominally distinct participants. Pooling, delegation, and related coordination mechanisms allow a single underlying concentration of control to appear as a large and diverse population of participants, obscuring the relationship between visible participation and effective control.

From this perspective, concentration is not merely a byproduct of specific protocol choices. It may emerge from the properties of the resource itself. The central question is therefore not only how influence is allocated, but whether the underlying resource permits influence concentration to be sustained at sublinear cost over time.

1.3 The Missing Question

The observation above points to a question that existing analyses rarely address:

> *What structural properties of a security resource determine whether sustained influence can be achieved at sublinear cost over time?*

Most deployed systems have approached concentration as a protocol-design problem. Mining-pool dominance, validator concentration, and staking monopolies are typically attributed to imperfect incentives, governance rules, or participant behavior. This perspective naturally motivates new protocol mechanisms intended to improve decentralization.

Our starting point is different. Rather than asking how a protocol allocates influence, we ask whether the underlying resource itself permits influence to be accumulated, redistributed, and reused over time. If so, concentration may persist across many different protocol designs because it is enabled by the structural properties of the resource.

We formalize this question through the adversarial cost function $C(s, T)$: the minimum expenditure required to sustain influence proportional to s independent participation units over T consecutive windows, whether through identity replication, delegation, or pooling. Two regimes are particularly important. When $C(s, T) = o(sT)$, influence can be sustained through resource amortization, allowing a concentrated resource base to support long-term influence at sublinear cost. When $C(s, T) = \Omega(sT)$, sustained influence requires cost that grows proportionally with both scale and time.

The central goal of this paper is to identify which resource properties lead to each regime. To do so, we develop an axiomatic taxonomy linking resource structure to the scaling behavior of $C(s, T)$. The resulting framework distinguishes resource classes that admit influence amortization from those that enforce linear cost, providing a resource-centric perspective on decentralization and Sybil resistance.

1.4 This Paper

This paper provides an axiomatic resource taxonomy for permissionless systems and establishes a sharp asymptotic separation between two resource classes.

The *negative half of the taxonomy* (Theorem 5.1) identifies a class of resource-mechanism pairs that admit *influence amortization*. Specifically, any resource-mechanism pair satisfying divisibility, additivity of influence, temporal reusability, and identity transferability yields $C(s,T) = o(sT)$. Mining hardware capacity, financial capital, storage, and delegated stake all fall into this class. (PoW energy, being window-local and non-reusable, does not satisfy temporal reusability and is analyzed separately as a flow component.)

The *positive half of the taxonomy* (Theorem 6.1) introduces *throughput-bounded resources*—in which each participation unit requires an independent channel, the channel output is non-transferable, and allocation does not carry over across time windows. We show that such resources enforce $C(s,T) = \Omega(sT)$: each additional unit of sustained influence incurs marginal cost $\Delta(s,T) = \Omega(T)$, growing with the time horizon.

Together, these two halves establish a sharp asymptotic separation: parallelizable resources admit influence concentration with reusable stock cost $O(s)$—the T-scaling renewal cost C_{flow} vanishes once stock is acquired—whereas throughput-bounded resources preclude such amortization.

The separation yields a direct design rule—the *Resource Substitution Theorem*: enforcing linear cost of influence concentration requires grounding participation in a resource that violates at least one parallelizability property. Within the scope of Definition 4.6, achieving this boundary requires altering at least one of the structural properties of the underlying resource.

1.5 Contributions

This paper develops an axiomatic framework for reasoning about influence concentration in permissionless systems. Rather than focusing on specific consensus protocols, we study how the structural properties of security resources shape the cost of sustaining influence over time. Our contributions are as follows.

(1) **An adversarial cost framework for influence concentration.**
We introduce the adversarial cost function $C(s,T)$, which measures the minimum expenditure required to sustain influence equivalent to s independent participants over a time horizon of length T. The framework separates one-time stock costs from recurring flow costs and provides a common basis for analyzing influence concentration across heterogeneous resource classes.

(2) **An axiomatic resource taxonomy for adversarial cost scaling.**
We identify a set of structural properties—divisibility, additivity of influence, temporal reusability, identity transferability, and throughput boundedness—that govern how adversarial cost scales over time. The taxonomy links resource structure directly to the asymptotic behavior of $C(s,T)$, allowing different resource classes to be compared within a common analytical framework.

(3) **A structural upper bound on influence cost under parallelizable resources.**
We prove that any resource-mechanism pair satisfying the parallelizability properties admits influence amortization, yielding $C(s,T) = o(sT)$. Consequently, a resource base of size $O(s)$ can sustain s-unit influence over an arbitrarily long horizon without costs that grow proportionally with time. The result depends only on the structural properties of the resource-mechanism pair and is independent of the specific consensus protocol built upon it.

(4) **A complementary linear-cost resource class and a design principle for decentralization.** We identify throughput-bounded, non-transferable, window-local resources as a class that enforces $C(s, T) = \Omega(sT)$ and establish a sharp asymptotic separation from parallelizable resources. This separation yields the *Resource Substitution Theorem*: achieving linear cost of influence concentration requires violating at least one of the structural properties associated with parallelizability. Within the scope of Definition 4.6, this boundary cannot be crossed while preserving all four parallelizability properties.

1.6 Paper Organization

Section 2 reviews related work. Section 3 presents the system model. Section 4 develops the resource framework. Section 5 proves the sublinear cost bound. Section 6 proves the linear lower bound. Section 7 establishes the structural separation and the Resource Substitution Theorem. Section 8 classifies concrete resource types. Section 9 illustrates the asymptotic separation. Section 10 discusses limitations. Section 11 concludes.

2 RELATED WORK

Prior work on Sybil resistance and permissionless systems addresses several distinct security questions. We survey the main research directions and identify the gap this paper fills. Table 1 summarizes the distinction.

2.1 Resource-Based Sybil Defenses

Douceur [12] established that in the absence of a trusted authority, identity replication in open systems cannot be prevented. Nakamoto [32] proposed binding participation to a scarce resource—computational effort—as a practical response to this impossibility. Formal analyses of Nakamoto-style consensus [16, 35, 36, 39] establish chain-growth, common-prefix, and chain-quality guarantees expressed as thresholds on adversarial hash power. Proof-of-Stake protocols—including Ouroboros [22–24], Snow White [37], and Algorand [20]—replace computational effort with financial capital and derive analogous guarantees as thresholds on adversarial stake. These protocols collectively instantiate the aggregate-threshold adversarial model formalized in foundational Byzantine fault-tolerance work [7, 26]; Algorand additionally employs verifiable random functions [30] for committee selection.

These analyses consistently reason about *aggregate* adversarial resource ownership. Security guarantees take the form: if the adversary controls less than a threshold fraction of the total resource, the protocol is safe. They do not characterize whether a stock of size $O(s)$ can sustain s-unit influence over T windows at cost $o(sT)$—that is, without T-proportional renewal. The present paper addresses this question directly.

2.2 Empirical and Systems-Level Analyses

Empirical work documents the centralization patterns that resource-based mechanisms produce in practice. Gencer et al. [17] measure mining pool concentration in Bitcoin and Ethereum. Rosenfeld [41, 42] analyzes pooled mining reward systems and hash-rate-based double-spending dynamics. Bonneau et al. [2] and Narayanan et al. [33] survey the broader security and deployment landscape of Bitcoin. Gervais et al. [18, 19] quantify security-performance tradeoffs and decentralization properties of deployed proof-of-work systems. Work on selfish mining [14] and double-spending [25] analyzes adversarial strategies under aggregate resource models. Network-layer studies [34] document how propagation dynamics interact with pool-driven resource concentration.

These works describe the *symptoms* of structural resource properties—concentration, pooling, delegation— but do not formally characterize why such patterns emerge as a structural consequence of the resource properties considered in this model. The present paper provides this characterization.

2.3 Economic Analyses of Consensus

Budish [4] and Budish et al. [5] analyze the economic limits of permissionless consensus, framing security in terms of the cost of mounting a majority attack. Carlsten et al. [6] study incentive stability in the absence of block rewards. Saleh [43] provides an equilibrium analysis of Proof-of-Stake.

These analyses model adversarial capability in terms of aggregate resource ownership and study equilibrium incentives or corruption costs. They do not study whether influence can be concentrated at sublinear cost through delegation or pooling—the question central to this paper.

2.4 Graph-Based and Social-Structure Defenses

SybilGuard [46], SybilLimit [47], and SybilControl [29] bound adversarial influence by exploiting the observation that in social networks, the number of edges between honest and Sybil regions is small relative to internal connectivity. SybilInfer [10] applies probabilistic inference over trust graphs to identify adversarial clusters. Game-theoretic analyses [27] study equilibrium conditions under which honest participants can deter adversarial identity creation.

These approaches address a different question: where Sybil identities can be placed within a social graph, rather than how the cost of adversarial influence scales over time. They provide no characterization of $C(s, T)$ as s or T grows, and their guarantees depend on graph-topology assumptions that may not hold in practice [45].

2.5 Identity-Binding Mechanisms

Proof-of-Personhood [3] and pseudonym parties [15] bind credentials to real-world participants to enforce one-person–one-identity mappings. Reputation systems [40] offer a related but distinct approach: rather than binding participation to verified identity, they tie influence to accumulated behavioral history. These mechanisms address *admission*—restricting the creation of new identities—rather than the sustained cost of concentrating influence over time. Once admitted, influence may still be governed by a transferable, reusable resource.

More recent deployed systems instantiate these admission mechanisms directly. World ID [44] binds a single credential to a unique biological human via biometric verification, and maps onto the *human real-time participation* class of Table 3: each verified human constitutes one per-actor channel. Privacy Pass [11] and its rate-limited extensions [8] bind anonymous credentials to per-user quotas, instantiating the *rate-limited participation* class—subject to the channel-binding condition of Definition 4.5. As with the device-bound and human-participation instantiations of Section 8, the structural guarantees apply only to the extent that the underlying credential cannot be resold, shared, or generated by automated agents.

2.6 Capability and Rate-Limiting Architectures

Hardware-rooted execution environments such as Intel SGX [9] enforce per-device participation limits that bind resource generation to specific hardware. Capability-based systems [28] and congestion-control mechanisms [21] impose per-actor throughput constraints in different deployment contexts. Bitcoin-NG [13] decouples block leadership from transaction throughput, instantiating a per-leader rate constraint as a protocol-level primitive. HoneyBadger

BFT [31] makes explicit throughput bounds central to its communication complexity analysis. Alwen et al. [1] study sustained space and time complexity of sequential proof-of-work constructions.

These works impose per-actor throughput constraints but analyze them in the context of access control, network stability, or computational hardness—not the cost of influence concentration. They do not establish the structural separation between parallelizable and throughput-bounded resource classes that this paper proves.

2.7 Positioning

The common thread across prior work is a focus on *aggregate* adversarial capability: how much resource does an adversary control, and is it below a safety threshold? To our knowledge, prior work does not ask whether a fixed aggregate resource can be used to concentrate influence at sublinear marginal cost—whether through identity replication, delegation, or pooling—nor establish the structural conditions under which this is or is not possible.

This paper provides that characterization. We identify the structural properties that determine whether adversarial cost scales as $o(sT)$ or $\Omega(sT)$, establish that parallelizable resources admit sublinear influence cost, introduce the throughput-bounded resource class and prove its linear cost guarantee, and derive a design principle for systems requiring linear cost of influence concentration.

3 SYSTEM MODEL

We model an open and permissionless system in which identity creation is inexpensive and no admission authority exists. Influence is regulated by a security-weighting resource $\mathcal{R}$. Our objective is to characterize how structural properties of $\mathcal{R}$ determine whether influence can be concentrated at sublinear cost—whether through identity replication, delegation, or pooling.

Table 2 summarizes the core notation.

3.1 Identities and Participation

An identity is a cryptographic key pair that authorizes participation. A single entity may control arbitrarily many identities at negligible cost [12], so influence may be acquired through *identity replication* or *resource aggregation*—the model captures both uniformly.

Let $S_t \subseteq \mathcal{V}_t$ denote the adversarial identities at window t, with $|S_t| = s_t$. The adversary $\mathcal{A}$ is *identity-unbounded*: it may create, retire, and coordinate an arbitrary number of identities at negligible syntactic cost.

3.2 Time Model

Time proceeds in discrete windows $t = 1, 2, \ldots$, representing the minimal interval over which resource allocation and influence are refreshed. Security is evaluated over a horizon of T windows.

3.3 Resource and Influence

For each identity $v \in \mathcal{V}_t$ and window t, let $r_v(t) \in \mathbb{R}_{\geq 0}$ denote the resource allocated to v. Influence is determined by a mapping

$$W_v(t) = f\big(r_v(t)\big), \tag{1}$$

where $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is the protocol's resource-to-influence function; structural assumptions on f are deferred to Section 4. The total adversarial influence at window t is $W_{\mathcal{A}}(t) = \sum_{v \in S_t} W_v(t)$.

Table 1. Positioning of This Work Relative to Prior Literature

Research Line	**Representative Works**	**Primary Security Question**	**Influence Conc. Cost?**
Sybil attack literature	[10, 12, 46, 47]	Can adversaries create multiple identities, and how can identity creation be restricted or detected?	No
Blockchain consensus security	[7, 16, 20, 24, 26, 39]	Under what aggregate adversarial resource ownership do safety and liveness properties fail?	No
Economic analyses of consensus	[4, 5, 25]	What equilibrium incentives and corruption costs arise in resource-weighted consensus?	No
Graph-based defenses	[10, 46, 47]	Can Sybil proliferation be constrained via social-network structure?	No
Identity-binding mechanisms	[3, 15]	How can systems enforce one-person–one-identity mappings?	No
Capability / rate-limiting	[9, 21, 28]	Can per-actor throughput limits be enforced at the system level?	No
This paper	—	How do structural properties of security resources determine whether influence can be concentrated at sublinear cost?	**Yes**

Definition 3.1 (Activation Threshold). *There exists a constant* $r_{\min} > 0$ *such that an identity is active in window* t *only if* $r_v(t) \geq r_{\min}$.

3.4 Adversarial Model

The adversary $\mathcal{A}$ is computationally unbounded but constrained solely by the structural properties of $\mathcal{R}$. It may instantiate, retire, and coordinate identities; allocate and reallocate resource; reuse resource across windows; and aggregate influence through delegation or pooling—whenever the resource structure permits.

3.5 Adversarial Cost

We distinguish between *resource allocation* and *new expenditure*, since the two differ for temporally reusable resources.

Definition 3.2 (Allocation and Expenditure). *For each identity* $v \in S_t$ *and window* t*:*

- $r_v(t) \geq 0$ *is the resource* allocated *to* v*—the amount drawn upon, whether freshly acquired or carried forward.*

- $e_v(t) \geq 0$ *is the resource* newly expended—*the incremental cost actually incurred in that window.*

For a reusable stock resource, $e_v(t) = 0$ for $t \geq 2$; for a window-local resource, $e_v(t) \geq r_{\min}$ must hold every window. Let $E_{\mathcal{A}}(t) = \sum_{v \in S_t} e_v(t)$.

Definition 3.3 (Adversarial Cost Function). *Let $W_{\text{unit}} = f(r_{\min})$. $C(s,T)$ denotes the minimum total* new expenditure *required to achieve influence equivalent to that of s independent participants over T consecutive windows:*

$$C(s,T) \;=\; \inf \left\{ A(\{r_v(1)\}) + \sum_{t=1}^{T} E_{\mathcal{A}}(t) + h(s,T) \;\middle|\; W_{\mathcal{A}}(t) \geq s \cdot W_{\text{unit}},\ \forall\, t \in \{1,\ldots,T\} \right\}, \tag{2}$$

where $A(\{r_v(1)\})$ is the one-time acquisition cost of the initial stock and $h(s,T)$ is the coordination overhead.

Because $C(s,T)$ is defined as an infimum over all admissible adversarial strategies, the model already captures adversaries that adapt identity allocation, resource distribution, and participation patterns over the horizon. The limitation discussed in Section 10.6 concerns richer dynamic models in which the resource-mechanism pair itself evolves in response to adversarial behavior.

The stock–flow distinction is central. For a reusable resource, $\sum E_{\mathcal{A}}(t) = 0$ once stock is acquired, so $C(s,T)$ grows only with s, not T. For a window-local resource, $\sum E_{\mathcal{A}}(t) \geq s \cdot T \cdot r_{\min}$, so $C(s,T)$ grows with both.

We decompose $C(s,T)$ as

$$C(s,T) = C_{\text{stock}}(s) + C_{\text{flow}}(s,T) + h(s,T), \tag{3}$$

where $C_{\text{stock}}(s) = A(\{r_v(1)\})$ is the one-time acquisition cost, $C_{\text{flow}}(s,T) = \sum_{t=1}^{T} E_{\mathcal{A}}(t)$ is the cumulative renewal expenditure, and $h(s,T)$ is coordination overhead.

Assumption 3.4 (Amortized Coordination). *The coordination overhead satisfies:*

(a) *$h(s,T) = o(sT)$; and*
(b) *$h(s,T) - h(s-1,T) = o(T)$ for each fixed s as $T \to \infty$.*

Both conditions are consistent with coordination infrastructure observed in practice, including mining pools, validator delegation services, and cloud-managed orchestration [2, 41, 43]. Condition (a) is used in Theorem 5.1; condition (b) bounds the marginal cost $\Delta(s,T)$.

3.6 Scaling Regimes

Definition 3.5 (Scaling Regimes). *A mechanism exhibits* sublinear scaling *if $C(s,T) = o(sT)$, and* linear scaling *if $C(s,T) = \Omega(sT)$.*

Under sublinear scaling, a resource base of size $O(s)$ can sustain influence equivalent to s participants over an arbitrarily long horizon.

Under linear scaling, sustaining influence equivalent to s participants over T windows requires cost growing as $\Omega(sT)$.

The central question of this paper is which scaling regime a given resource $\mathcal{R}$ induces. Section 4 provides the structural answer.

Table 2. Core Notation

Symbol	Meaning
v	Identity (key pair / participant instance)
t	Time window index
T	Time horizon (number of windows)
s	Influence level in participant equivalents
$r_v(t)$	Resource allocated to identity v at window t
$e_v(t)$	New expenditure by identity v in window t
$R_{\mathcal{A}}(t)$	Total adversarial resource allocated at window t
$E_{\mathcal{A}}(t)$	Total new adversarial expenditure in window t
$W_v(t)$	Influence of identity v at window t
W_{unit}	Baseline influence: $f(r_{\min})$
$f(\cdot)$	Resource-to-influence mapping
S_t	Set of adversarial identities at window t
$C(s,T)$	Min. cost to achieve influence equivalent to s participant equivalents over T windows
$\Delta(s,T)$	Marginal cost: $C(s,T) - C(s-1,T)$
$r_{\min}$	Activation threshold
τ	Per-channel throughput bound
$h(s,T)$	Coordination overhead

4 RESOURCE FRAMEWORK

We characterize security-weighting resources by their structural properties—the constraints they impose on division, aggregation, reuse, and transfer across identities and time. These properties play a central role in determining whether influence concentration can be sustained at sublinear cost.

4.1 Structural Properties

Definition 4.1 (Divisibility). $\mathcal{R}$ is divisible *if for any allocation $r \geq 0$ and integer $k \geq 1$, there exist $r_1, \ldots, r_k \geq 0$ with $\sum_{i=1}^{k} r_i = r$, and the adversary may distribute r_i to identity i without additional structural cost.*

Definition 4.2 (Additivity of Influence). *A mechanism* induces additive influence *if*

$$\sum_{v \in S_t} f\big(r_v(t)\big) \;=\; f\Bigg(\sum_{v \in S_t} r_v(t)\Bigg). \tag{4}$$

Total influence depends only on aggregate allocation, not on how it is partitioned across identities.

Remark 4.1. Definition 4.2 is a property of the influence function f, not of the resource alone. Mechanisms that deliberately penalize concentration— such as quadratic voting, where $f(r) = \sqrt{r}$— do not satisfy it and fall outside the scope of Theorem 5.1. Theorem 5.1 applies only to resource-mechanism pairs for which influence is additive in this sense; non-additive resource-to-influence mappings may induce different cost-scaling behavior. Characterizing the scaling regimes admitted by non-additive mechanisms is outside the scope of the present work.

Additivity is not posited as a universal property of resource-to-influence mappings, but as the property that characterizes the influence mechanisms underlying the major resource-weighted systems motivating this work—including Proof-of-Work mining power, stake-weighted participation, delegated stake, and pooled resource ownership. Theorem 5.1 therefore characterizes this practically dominant class of resource-mechanism pairs; mechanisms with non-additive influence functions are addressed by the scope limitation above.

Definition 4.3 (Temporal Reusability). *$\mathcal{R}$ is* temporally reusable *if an allocation obtained in window t may be applied in window $t + 1$ without renewed identity-bound expenditure.*

Definition 4.4 (Identity Transferability). *$\mathcal{R}$ is* identity-transferable *if an allocation $r_v(t)$ assigned to identity v may be reassigned to any other identity v' without proportional cost.*

Definition 4.5 (Throughput-Bounded Resource). *$\mathcal{R}$ is* throughput-bounded *if each identity v is associated with a dedicated participation channel C_v satisfying:*

(1) Per-channel rate limit. *There exists $\tau > 0$ such that $r_v(t) \leq \tau$ for all v and t.*
(2) Non-transferability. *$r_v(t)$ cannot be simultaneously credited to any other identity $v' \neq v$.*
(3) Window-locality. *Allocation in window t does not carry over to window $t + 1$; sustained participation requires renewed expenditure in every window.*

We assume $0 < r_{\min} \leq \tau$ throughout.

4.2 Resource Classes

Definition 4.6 (Parallelizable Resource). *$\mathcal{R}$ is* parallelizable *if it satisfies divisibility, temporal reusability, and identity transferability, and the induced mechanism satisfies additivity of influence.*

Definition 4.7 (Throughput-Bounded Non-Parallelizable Resource). *$\mathcal{R}$ is a* throughput-bounded non-parallelizable resource *if it satisfies Definition 4.5.*

4.3 Resource Taxonomy

Table 3 classifies representative resource types by the structural properties they satisfy. Proof-of-Work is decomposed into its capacity component (mining hardware: parallelizable stock) and its operational component (energy: window-local flow).

5 SUBLINEAR COST UNDER PARALLELIZABLE RESOURCES

We prove that no resource-mechanism pair satisfying the four parallelizability properties can enforce linear cost of influence concentration.

Theorem 5.1 (Amortization Under Parallelizable Resources). *Let $(\mathcal{R}, f)$ be a parallelizable resource-mechanism pair (Definition 4.6). Under Assumption 3.4,*

$$C(s, T) = o(sT). \tag{5}$$

In particular, $\Delta(s, T) = C(s, T) - C(s - 1, T) = o(T)$: the marginal cost of influence concentration is sublinear in the time horizon.

Proof. By divisibility and additivity of influence, any aggregate allocation r may be partitioned across s identities without changing total influence: $\sum_v f(r_v) = f(r)$ for any partition $\{r_v\}$ with $\sum_v r_v = r$. By temporal reusability, an

Table 3. Structural Taxonomy of Security Resources. PoW is decomposed into its capacity component (hardware: parallelizable) and operational component (energy: non-reusable flow). **Add.** indicates whether the mechanism induces additive influence (Definition 4.2).

Resource	Div.	Add.	Reuse	Transfer	Notes	Scaling
PoW: hardware (capacity)	✓	✓	✓	✓	Parallelizable stock; mining pools instantiate amortization	$o(sT)$
PoW: energy (operational)	✓	✓	×	✓	Window-local flow; renewed each block interval	Linear in $T^{\dagger}$
Financial stake (PoS)	✓	✓	✓	✓	Parallelizable stock; delegation markets instantiate amortization	$o(sT)$
Social-graph trust	Partial	Partial	Partial	Partial	Guarantees depend on graph topology; no explicit $C(s,T)$ bound	Model-dep.
Device-bound execution	×	×	×	×	Each device constitutes one independent channel per window	$\Omega(sT)$
Human real-time participation	×	×	×	×	Cognitive and temporal throughput limits bound channel capacity	$\Omega(sT)$
Rate-limited participation[‡]	×	×	×	×	Qualifies only when bound to non-transferable execution channels	$\Omega(sT)$

[†]Energy is non-reusable but transferable; it falls outside the parallelizable class of Theorem 5.1.

[‡]Account-level rate limits without channel binding are bypassable through identity replication.

allocation $r_v(1) = r_{\min}$ acquired in window 1 remains available in all subsequent windows without renewed expenditure. Therefore the adversary achieves s-unit influence at total cost

$$C(s,T) \;\leq\; s \cdot r_{\min} + h(s,T), \tag{6}$$

where $C_{\text{flow}}(s,T) = 0$ since no renewal expenditure is required after window 1. By Assumption 3.4(a), $h(s,T) = o(sT)$, so

$$\frac{C(s,T)}{sT} \leq \frac{r_{\min}}{T} + \frac{h(s,T)}{sT} \longrightarrow 0 \quad \text{as } s,T \to \infty. \tag{7}$$

From (6),

$$\Delta(s,T) \;\leq\; r_{\min} + h(s,T) - h(s-1,T). \tag{8}$$

By Assumption 3.4(b), $h(s,T) - h(s-1,T) = o(T)$ for each fixed s, so $\Delta(s,T) = o(T)$. □

The result is structural: for any parallelizable resource-mechanism pair, the T-scaling flow cost $C_{\text{flow}}(s,T)$ vanishes once the initial stock is acquired, so the reusable resource component requires only $O(s)$ stock cost; any additional dependence on the time horizon arises through coordination overhead $h(s,T)$ rather than resource renewal. Enforcing linear cost therefore requires violating at least one of the parallelizability properties, whether through the choice of resource or through protocol mechanisms that prevent those properties from being realized in practice.

6 LINEAR LOWER BOUND UNDER THROUGHPUT-BOUNDED RESOURCES

We prove the complementary result: throughput-bounded non-parallelizable resources structurally enforce $C(s,T) = \Omega(sT)$.

Theorem 6.1 (Linear Lower Bound). *Let $\mathcal{R}$ be a throughput-bounded non-parallelizable resource (Definition 4.7). Then there exists $c > 0$ such that*

$$C(s,T) \;\geq\; c \cdot sT, \tag{9}$$

and in particular $C(s,T) = \Omega(sT)$.

Proof. Set $c = r_{\min}$. By non-transferability (condition 2 of Definition 4.5), each active identity requires its own allocation: $r_v(t) \geq r_{\min}$ must hold independently for each of the s identities. By window-locality (condition 3), allocation does not carry forward, so new expenditure equals allocation in every window. Therefore $E_{\mathcal{A}}(t) \geq s \cdot r_{\min}$ for every t, and

$$C(s,T) \;\geq\; \sum_{t=1}^{T} E_{\mathcal{A}}(t) \;\geq\; s \cdot T \cdot r_{\min} \;=\; c \cdot sT. \tag{10}$$

□

Unlike parallelizable resources, expenditure cannot be amortized across identities or across time. Each active identity requires renewed expenditure in every window, causing both the number of participants and the time horizon to contribute linearly to cost.

7 STRUCTURAL SEPARATION AND DESIGN CONSEQUENCES

Theorems 5.1 and 6.1 characterize two fundamentally different classes of security-weighting resources. We now combine these results into a single structural separation and derive its implications for mechanism design.

7.1 Structural Separation

Theorem 7.1 (Structural Separation). *Let $(\mathcal{R}, f)$ be a security-weighting resource-mechanism pair.*

(1) *If $\mathcal{R}$ is parallelizable, then $C(s,T) = o(sT)$.*
(2) *If $\mathcal{R}$ is a throughput-bounded non-parallelizable resource, then $C(s,T) = \Omega(sT)$.*

The two classes induce asymptotically disjoint scaling regimes.

Proof. Part (1) is Theorem 5.1. Part (2) is Theorem 6.1. Since $o(sT)$ and $\Omega(sT)$ are asymptotically disjoint, no resource can exhibit both regimes. □

Theorem 7.1 establishes that the distinction between amortizable and non-amortizable influence is not quantitative but structural. Parallelizable resources admit sublinear scaling, whereas throughput-bounded non-parallelizable resources enforce linear scaling. No resource satisfying one class definition can simultaneously exhibit the asymptotic behavior of the other.

7.2 Resource Substitution Theorem

Theorem 7.2 (Resource Substitution Theorem). *Any mechanism targeting $C(s,T) = \Omega(sT)$ must ground participation in a resource that violates at least one of the following properties: divisibility, additivity of influence, temporal reusability, identity transferability.*

Proof. If the underlying resource satisfies all four properties it is parallelizable (Definition 4.6), and Theorem 5.1 gives $C(s,T) = o(sT)$, a contradiction. □

Corollary 7.3. *For any resource-mechanism pair satisfying Definition 4.6 and Assumption 3.4, Theorem 5.1 implies that no protocol-level modification preserving these properties can enforce $C(s,T) = \Omega(sT)$. Enforcing linear cost therefore requires violating at least one parallelizability property.*

The Resource Substitution Theorem is the primary design consequence of this work. Within the class of parallelizable resources, linear influence cost cannot be achieved through protocol engineering alone. Any mechanism seeking $C(s,T) = \Omega(sT)$ must ultimately rely on a resource that violates at least one of the four parallelizability properties. The boundary between amortizable and non-amortizable influence is therefore determined by resource structure rather than by protocol rules.

8 INSTANTIATIONS AND CLASSIFICATION

This section instantiates the framework by classifying representative resource types according to the structural properties of Definitions 4.1–4.5.

8.1 Per-Actor Throughput Channels

Definition 8.1 (Per-Actor Throughput Channel). *A participation channel C_u associated with actor u is a* per-actor throughput channel *if there exists $\tau > 0$ such that for every window t, $\mathrm{rate}_t(C_u) \leq \tau$, and $r_v(t) \leq \mathrm{rate}_t(C_u)$ for any identity v backed by C_u.*

Proposition 8.2 (Per-Actor Channel Classification). *Any resource derived from per-actor throughput channels is a throughput-bounded non-parallelizable resource (Definition 4.7). Consequently, $C(s,T) = \Omega(sT)$ by Theorem 6.1.*

Proof. Throughput boundedness follows from the rate limit τ of Definition 8.1. Non-transferability holds because C_u's output cannot be credited to an identity under a different actor. Window-locality holds because $\mathrm{rate}_t(C_u)$ does not carry over to window $t+1$. Therefore Definition 4.5 is satisfied, and the resource belongs to the throughput-bounded non-parallelizable class. □

8.2 Classification of Deployed Resources

Proof-of-Work (hardware component). Mining hardware is divisible, transferable, temporally reusable, and induces additive influence. It therefore satisfies the parallelizable resource class, implying $C(s,T) = o(sT)$ by Theorem 5.1. Mining pools provide an empirical manifestation of this amortization [17, 41].

Proof-of-Stake (financial capital). Financial stake satisfies the same four parallelizability properties. It therefore induces $C(s,T) = o(sT)$ by Theorem 5.1. Large staking pools and delegation systems provide empirical manifestations of this amortization pattern.

Device-bound execution. Each trusted execution environment constitutes one per-actor throughput channel. By Proposition 8.2, $C(s,T) = \Omega(sT)$, subject to the deployment fidelity conditions discussed in Section 10.1. The classification is structural: it characterizes the resource as defined by Definition 4.5, not any particular hardware implementation. In practice, SGX-like environments may deviate from this idealization through relay attacks, remote attestation by a third party, virtualization of the enclave, or leasing and outsourcing of attestation capacity to other actors—each of which can weaken non-transferability or window-locality. To the extent that such deviations are absent, the resource approximates a throughput-bounded channel under the framework's assumptions; the consequences of partial deviation are discussed further in Section 10.

Human real-time participation. Each participant constitutes one channel with intrinsic per-window throughput limits. By Proposition 8.2, $C(s,T) = \Omega(sT)$, subject to the same deployment fidelity conditions. As with device-bound execution, this classification is structural and reflects the idealized properties of Definition 4.5 rather than any specific deployment. Deployed human-participation systems may depart from this idealization through credential markets that resell verified identities, outsourcing of participation tasks to third parties, or automation that mimics human responses—each of which can erode non-transferability or the per-channel rate limit. Provided such practices remain limited, the resource approximates a throughput-bounded channel under the framework's assumptions; Section 10 discusses the open problem of characterizing this approximation more precisely.

Rate-limited network participation. Account-level rate limits qualify as throughput-bounded only when bound to non-transferable execution channels. Otherwise, limits attached solely to syntactic identities can be circumvented through identity replication.

9 ILLUSTRATING THE STRUCTURAL SEPARATION

The following figures illustrate the asymptotic behavior implied by Theorems 5.1 and 6.1 for a representative coordination overhead $h(s,T) = s + T$ satisfying Assumption 3.4; they are not empirical measurements. With this choice,

$$\frac{C_{\text{par}}(s,T)}{sT} = \frac{r_{\min}+1}{T} + \frac{1}{s},$$

which vanishes only in the joint limit $s, T \to \infty$, not along fixed-s or fixed-T slices.

9.1 Cost Scaling and Joint-Asymptotic Convergence

Figure 1 shows the normalised ratio along the diagonal $s = T$. Along this diagonal,

$$\frac{C_{\text{par}}(s,s)}{s^2} = \frac{r_{\min}+2}{s} \to 0 \qquad \text{as } s \to \infty,$$

confirming Theorem 5.1 in the joint regime. The throughput-bounded ratio remains fixed at $r_{\min}$ (Theorem 6.1).

9.2 Non-Amplification Under Identity Replication

Under a throughput-bounded resource, influence is determined by channel capacity, not identity count. An adversary controlling m channels and $s \in \{400, 700, 1000\}$ identities against $n = 200$ honest validators achieves influence share $m/(m+n)$, independently of s. Since $s > m$ throughout, all three curves in Figure 2 overlap exactly: multiplying adversarial identities by 2.5 yields zero influence gain, confirming the non-amplification consequence of Theorem 6.1.

10 LIMITATIONS AND OPEN PROBLEMS

The structural separation established in this paper characterizes adversarial cost scaling under explicit modeling assumptions. Several limitations and open directions remain.

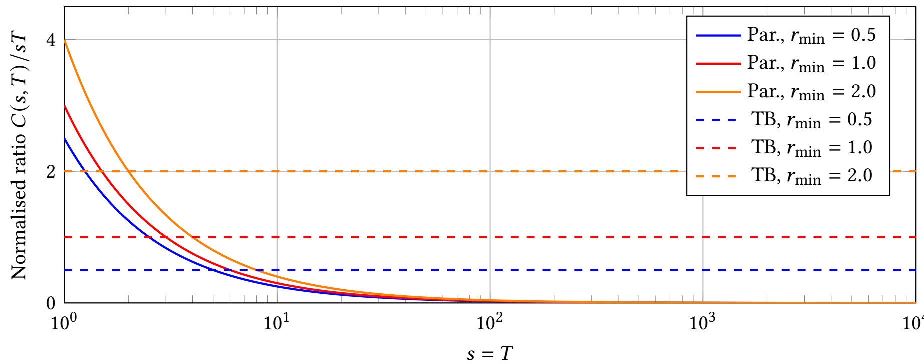

Fig. 1. Normalised ratio $C(s,T)/sT$ along the joint diagonal $s = T$. Solid curves satisfy $(r_{min} + 2)/s \to 0$, illustrating the asymptotic behavior established by Theorem 5.1. Dashed lines remain fixed at r_{min}, illustrating the linear-cost regime of Theorem 6.1.

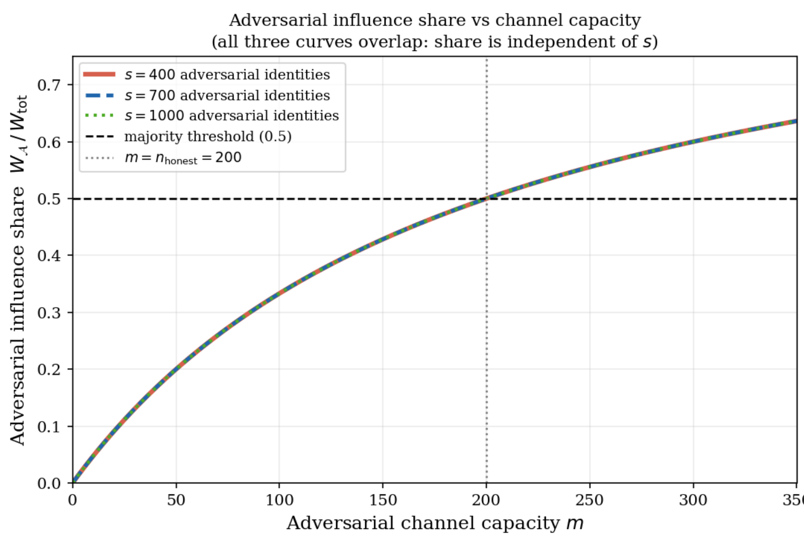

Fig. 2. Adversarial influence share vs. channel capacity m, for $s \in \{400, 700, 1000\}$ and $n_{honest} = 200$. All curves overlap: influence share equals $m/(m+n)$, independent of s. Identity replication alone cannot increase influence; additional influence requires additional throughput channels (Theorem 6.1).

10.1 Enforcement of Throughput Constraints

The framework assumes that the structural properties of Definition 4.5—non-transferability, window-locality, and per-channel rate limits—hold in the deployment. In practice, mechanisms may only approximate these properties: device attestations can be rented or remotely operated, human participation can be outsourced, and advances in automation may effectively increase τ, weakening the per-channel rate constraint. Characterizing the degradation of the $C(s,T) = \Omega(sT)$ guarantee as a function of deployment-level fidelity remains an open problem.

10.2 Non-Additive Hybrid Compositions

The linear lower bound of Theorem 6.1 is established under an additive cost decomposition $C(s,T) = C_p(s,T) + C_t(s,T)$, where a throughput-bounded component gates all influence units. Two open directions remain. First, non-additive compositions—where the two resource classes interact in ways not captured by additive decomposition—fall outside the present scope; it is not yet known whether the linear lower bound survives under such interactions. Second, partial gating arrangements, in which a throughput-bounded component constrains only a fraction β of influence units, may yield intermediate scaling regimes of the form $C(s,T) = \Omega(\beta \cdot sT)$, but a general characterization remains open.

10.3 Economic Coordination and Identity Markets

The model abstracts away economic coordination dynamics. In practice, delegation markets, identity leasing, credential outsourcing, and secondary markets for rate-limited participation rights may alter the effective adversarial cost structure even when the underlying structural resource properties remain unchanged. An adversary with access to liquid identity markets may, for instance, effectively circumvent non-transferability constraints by acquiring participation slots through market intermediaries rather than through direct channel ownership. Integrating equilibrium analysis—capturing strategic outsourcing, adaptive pricing, and rational credential supply—with the structural framework developed here is an important direction for future research.

10.4 Scope of Assumption 3.4

Assumption 3.4 provides a sufficient condition for the sublinear cost result of Theorem 5.1. Its purpose is not to model all conceivable coordination structures, but to isolate the contribution of resource reusability and transferability from coordination-specific effects. It reflects the coordination overhead observed in deployed infrastructures such as mining pools and validator delegation services, both of which exhibit coordination costs consistent with $h(s,T) = O(s+T)$ [41, 43]. The open question is not whether the theorem extends to overheads that remain $o(sT)$—Theorem 5.1 already covers that class—but rather how to realistically characterize coordination processes in practice: what functional forms of $h(s,T)$ arise in deployed systems, and how marginal-cost behavior $\Delta(s,T)$ varies under realistic coordination assumptions beyond the $O(s+T)$ benchmark.

10.5 Window Granularity and Temporal Modeling

The model partitions time into discrete windows of fixed duration. Smaller windows increase renewal frequency under throughput-bounded resources; larger windows may weaken window-local constraints. A window-independent formulation expressing the separation in terms of wall-clock time remains an open direction.

10.6 Scope of the Framework

The framework is structural and protocol-agnostic: it characterizes the resource-level preconditions that any mechanism must satisfy, and does not replace deployment-specific security analyses, probabilistic consensus proofs, or economic equilibrium models. Characterizing intermediate scaling regimes—where structural properties are only partially satisfied—in full generality remains an open theoretical problem. The model is also resource-centric and static: it characterizes the cost $C(s,T)$ induced by a fixed resource-mechanism pair, but does not model settings in which the pair $(\mathcal{R}, f)$ itself evolves over time—for example, through protocol updates made in response to observed adversarial behavior. Extending the framework to such dynamic resource-mechanism pairs is an important direction for future work.

11 CONCLUSION

Binding influence to a scarce resource prevents cheap identity creation from amplifying adversarial power— but scarcity alone is insufficient. The structural properties of the resource-mechanism pair determine whether influence cost is amortizable over time or necessarily recurring.

This paper provides an axiomatic resource taxonomy connecting structural properties to adversarial cost scaling $C(s,T)$. The negative half of the taxonomy (Theorem 5.1) establishes that any resource-mechanism pair satisfying divisibility, additivity of influence, temporal reusability, and identity transferability admits influence amortization: $C(s,T) = o(sT)$, with marginal cost $\Delta(s,T) = o(T)$. The positive half (Theorem 6.1) identifies throughput-bounded, non-transferable, window-local resources as a class that provably enforces $C(s,T) = \Omega(sT)$, with marginal cost $\Delta(s,T) = \Omega(T)$. Theorems 5.1 and 6.1 together establish a sharp asymptotic separation between the two resource classes (Theorem 7.1).

The Resource Substitution Theorem (Theorem 7.2) translates this separation into a design rule: enforcing linear cost of influence concentration requires grounding participation in a resource that violates at least one parallelizability property. Within the scope of Definition 4.6, no protocol rule can enforce this boundary without a structural change in the underlying resource.

More broadly, these results suggest that within mechanisms preserving the relevant parallelizability properties, decentralization limits arise primarily from resource structure rather than consensus rules alone. The structural properties of the underlying resource constitute a first-order design choice that shapes the long-term economics of influence concentration.

Future directions include the formal characterization of intermediate scaling regimes under partial violations of parallelizability, equilibrium analysis under adaptive incentives and identity markets, and the design of practical mechanisms that enforce throughput-boundedness while preserving permissionlessness and accessibility.

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